Chapter 23

Short-Dated Options, Dealer Positioning, and Market Feedback

Every chapter so far has treated the option market as a place where risk is transferred. This one is about what happens when the transfer becomes large enough, and fast enough, that the hedging it requires becomes a force in the underlying market.

The occasion is a market structure that did not exist when the previous generation of textbooks was written. Options expiring the same day they are traded, the 0DTE options of Section 10.12, went from a curiosity to the majority of S&P 500 option volume in under a decade. Section 10.12 recorded the growth and deferred the mechanism here; Section 18.12 supplied the machinery, a dealer whose hedging moves the market it is hedging in; and Section 21.12 observed that this market lives where classical stochastic volatility cannot make a skew.

Three demands meet in the contract: a hedger whose risk lasts one session (a data release, a policy decision) and who used to buy a month of protection for it, a retail buyer wanting a leveraged intraday bet, and a systematic seller collecting the volatility risk premium of Section 19.10 at the shortest horizon there is (Section 23.13). Whether the hedging of their positions stabilizes or destabilizes the index depends on how the three net.

The chapter has an unusual shape, because the answer to its central question is that we do not yet know. It first calculates the mechanism by which 0DTE could destabilize the market (Sections 23.2 to 23.11), then weighs the evidence, which so far says mostly no, and why that is not reassuring (Section 23.12), and ends with what would have to go wrong (Section 23.14).

23.1 WHAT 0DTE IS, AND HOW IT GREW

An option expiring on the day it trades is not new—monthly and quarterly expirations have always had a final day. What is new is that there is one *every* day.

S&P 500 index options (SPX: European, settled in cash, $100 per index point, the weekly and daily series against the closing level) acquired weekly expirations progressively: Fridays first, then Wednesdays and Mondays in 2016, Tuesdays and Thursdays in 2022. Since then every trading day has been somebody's expiration, and a trader can put on a same-day position at any time.

The share of volume moved with it. On Cboe's figures, 0DTE contracts were a single-digit percentage of SPX option volume in the mid-2010s, passed half at the end of 2024, and ran close to three-fifths through 2025. The most heavily traded S&P 500 option is now one that will not exist tomorrow. Two things about that sentence are worth separating.

Volume is not risk transfer. A 0DTE contract has a fraction of the vega of a monthly option and lives for hours rather than weeks, so measuring the market in contracts overstates how much exposure has changed hands. That overstatement is the most common error in commentary on the subject.

But volume is hedging. Every contract traded is hedged by somebody, and the hedging of a 0DTE contract is concentrated into hours rather than spread over months. The flow is

compressed even where the risk is not, and compression is what makes a market impact.

23.2 WHY SHORT-DATED OPTIONS BEHAVE DIFFERENTLY

Everything peculiar about this market follows from what happens to the Greeks as $T \to 0$. Take the S&P 500 at 5000 with a 15% implied volatility (a normal quiet market), with a 4% rate and a 1.5% dividend yield. Time is counted in trading hours throughout, 6.5 to the day and 252 days to the year, because that is the clock a 0DTE option lives on; every number in the chapter is priced this way, and "30 days" means thirty trading days, about six calendar weeks.

Table 23.1: At-the-money call as expiration approaches. The last two columns are the delta-hedged profit per unit of the index on a 1% rise.

Time left	Price	Delta	Gamma	Gamma term	Exact P&L
30 days	110.47	0.5323	0.001534	1.92	1.89
7 days	51.58	0.5158	0.003188	3.98	3.90
1 day	19.10	0.5060	0.008443	10.55	9.60
4 hours	14.94	0.5047	0.010763	13.45	11.65
1 hour	7.43	0.5024	0.021528	26.91	17.55
15 minutes	3.71	0.5012	0.043056	53.82	21.25

Gamma at one hour is fourteen times gamma at thirty days (Table 23.1), and it keeps rising: at-the-money gamma diverges like $1/\sqrt{T}$, which Section 18.3's formula makes obvious once one looks at it. That one number gives the scale of the problem.

But the more important fact is not how large the gamma is—it is how *narrow* it is (Figure 23.1). The table's last two columns show it. Both are the hedged profit on a 1% rise, once as the gamma term $\frac{1}{2}\Gamma\,(0.01\,S)^2$ of the Taylor expansion and once by repricing the option; at one hour a 1% move is 2.7 hourly standard deviations, and gamma falls away as the index leaves the strike, to 3% of its at-the-money value by the end of the move. The exact profit is earned on a gamma that falls as the index moves, while the gamma term keeps the starting value, so the profit comes out a third below the gamma term, and at fifteen minutes it is two-fifths of it.

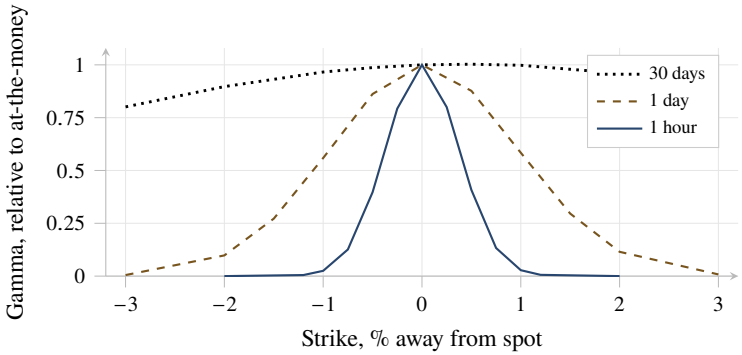


Figure 23.1: Gamma by strike, normalized to the at-the-money value.

Call the range of strikes over which gamma exceeds half its at-the-money value the **gamma band**. Its edges, which are symmetric in the logarithm of the strike about a point slightly above spot rather than about spot itself, close in as follows:

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Time to expiration	Edges of the gamma band, above / below spot
30 days	+6.76%/−5.52%
1 day	+1.13%/−1.09%
1 hour	+0.44%/−0.43%

This is the structural fact of the chapter. A month-dated book has gamma spread across a wide range of index levels, so a move of one percent changes the hedging requirement gradually. In its last hour a 0DTE book has four-fifths of its gamma within half a percent of spot and nearly all of it within one percent, so the hedging requirement is a step function of where the index happens to be. Move the index one percent and a dealer's entire gamma exposure can switch off, or switch on, or change sign.

23.3 THE BUYER'S ARITHMETIC

Before the market-structure question comes the individual trade. A 0DTE option is almost pure time value, and it has hours to live. Figure 23.2 traces that life through one session: with the index unchanged, the option loses value steadily for most of the day and then falls away in the final hour.

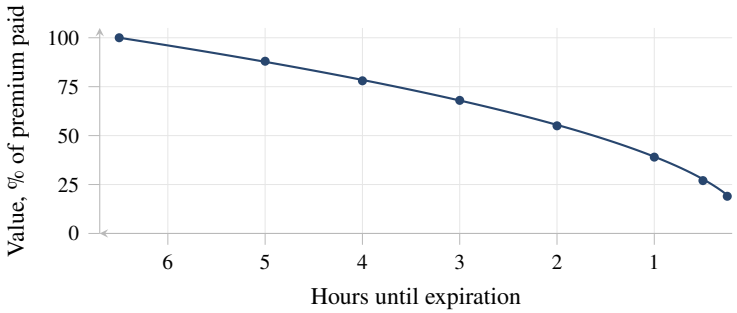


Figure 23.2: Value of an at-the-money 0DTE call through the session, with the index unchanged. Time runs left to right.

Example 23.1 What the buyer faces

An at-the-money 0DTE call on the index at 5000, bought at the open, costs 19.10, or **0.38% of spot**. A one-standard-deviation daily move is 47 points, or 0.94%.

The table gives the option's value through the day if the index does not move at all, and the rise the index needs for the buyer to break even on selling at each point:

Time left	Value, index unchanged	% of premium paid	Break-even rise
6.5 hours	19.10	100%	—
4 hours	14.94	78%	+0.15%
2 hours	10.53	55%	+0.28%
1 hour	7.43	39%	+0.35%
15 minutes	3.71	19%	+0.38%
Expiration	0	0%	+0.38%

An earlier exit needs a smaller rise, because the option is sold with time value still in it: with four hours left, 11.47 of the 19.10 recovered is time value.

Held to expiration, the buyer needs a 0.38% move in the right direction merely to get their money back, against a daily standard deviation of 0.94%. That is a hurdle of about 0.4 standard deviations before any profit. The option is not a bad bet in the way a lottery ticket is: it is priced

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without obvious error, with a negative expected return set mainly by the volatility risk premium of Section 19.10.

What makes it feel different is the *shape*. Sixty-one percent of the premium is gone by the last hour if nothing happens, so the position is either right quickly or worthless. The instrument converts a modest edge against the holder into a rapid and highly visible sequence of total wins and total losses, which is a design that human beings find much more engaging than a slow loss. Section 23.13 returns to that.

Note also what this does to bid–ask cost. At 19.10 a one-tick spread of ten cents (SPX's minimum tick for an option priced at $3 or more) is 0.52% of the premium; on the same option with fifteen minutes left, priced at 3.71, the same dime is 2.7%. Trading the same contract later in the day costs several times as much in percentage terms, and the volume is heaviest late.

23.4 CHARM: THE GREEK THAT ONLY MATTERS HERE

Section 18.6 named charm, the rate at which delta changes with the passage of time, and moved on, because for a monthly option it is a rounding error. Writing f for the option's value,

$$\text{charm} = \frac{\partial \Delta}{\partial t} = \frac{\partial^2 f}{\partial S \, \partial t}$$

with t the time elapsed. Charm is the cross-derivative that pairs with gamma: gamma is how the hedge changes when the price moves, charm is how it changes when the clock moves and the price does not. It is a cross term in $\Delta S \, \Delta t$ that Chapter 18's expansion, equation (18.1), leaves out, because for a monthly option the delta barely drifts with the clock alone. For 0DTE it is the dominant flow.

The reason is visible in Figure 23.1. As expiration approaches, an out-of-the-money option's delta must go to zero and an in-the-money option's to one. The whole delta surface is collapsing into a step, and that collapse happens on the clock rather than on the price. Figure 23.3 shows it on a single contract: with the index unchanged from open to close, the delta of a call struck just above spot falls the whole way down, and every point of that fall is a hedge somebody has to trade.

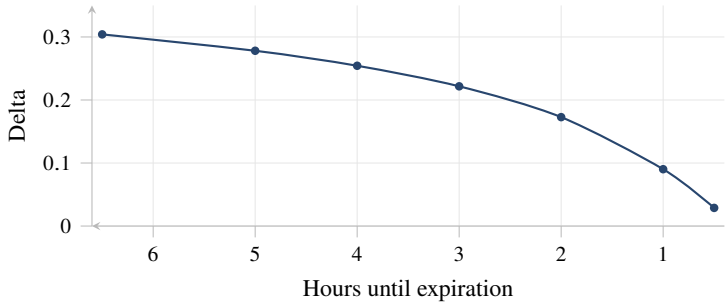


Figure 23.3: Delta of a call struck 0.5% above spot, with the index unchanged all day. Time runs left to right.

Example 23.2 A billion dollars of hedging, with the index unmoved

Take a call struck 0.5% above spot (about half a daily standard deviation, an ordinary strike for a 0DTE trade). With the index held at 5000 all day:

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Time left	Delta	Charm, per hour
6.5 hours	0.3041	−0.0146
4 hours	0.2543	−0.0274
2 hours	0.1728	−0.0614
1 hour	0.0902	−0.1099
0.5 hours	0.0288	−0.1254

A dealer short 10,000 of these contracts (index multiplier 100) hedges by holding 304,095 index units long at the open ($1.52 billion) and 90,164 units with one hour left, or $451 million. The dealer must *sell* roughly $1.07 billion of index by the last hour, with the index unmoved. Three properties distinguish this flow from the gamma hedging of Section 18.12.

It is one-directional. Gamma hedging buys on the way up and sells on the way down, netting out over a round trip. Charm hedging runs one way all day, set by the clock.

It is predictable. Anyone who knows the open interest by strike can compute it in advance, because the only input is the passage of time.

And it accelerates. The size of charm rises from 0.015 to 0.125 per hour, more than eightfold, so the flow is fastest in the final hour, which is also when 0DTE volume is heaviest and the day's other closing trades are competing for the same buyers and sellers.

Whether this stabilizes or destabilizes depends on the sign of dealer positioning, which Section 23.7 takes up. What is certain is that a large, predictable, clock-driven flow now exists in the final hour of the US equity session, and that ten years ago it was a fraction of its size.

23.5 THE OVERNIGHT GAP IS NOT IN THE CONTRACT

A quiet measurement problem sits underneath every comparison of 0DTE implied volatility with anything else, and it accounts for a discrepancy that is easily misread. A 0DTE option bought at the open expires at the close. It covers the session and not the night, yet it is quoted with a full trading day as its time to expiration. If the overnight period (close to next open) carries a fraction ω of daily variance, then the session carries $1 - \omega$, and the implied volatility that reproduces the option's price is

$$\sigma_{\text{session}} = \sigma_{\text{daily}} \sqrt{1 - \omega}$$

where σ_{daily} is the annualized volatility of close-to-close returns, the figure a 30-day implied volatility measures.

Overnight share ω	Implied volatility shown	Against 15% close-to-close
0%	15.00%	—
10%	14.23%	−5.1%
20%	13.42%	−10.6%
30%	12.55%	−16.3%
40%	11.62%	−22.5%

For a US index, where most news arrives during the session, ω is typically in the 15–25% range, so on the trading-day clock used here 0DTE implied volatilities print about a tenth below the 30-day implied volatility for purely mechanical reasons. (A calendar-minute clock makes the opposite error, showing 0DTE volatility well above the daily figure.) Cboe's one-day volatility index, VIX1D, launched in April 2023, publishes this session-only volatility, and comparing it with the 30-day VIX needs the same adjustment.

The lower print can be misread as evidence that short-dated options are cheap. They are not; the measurement window is shorter than the quoted maturity. Section 19.6's term structure is a real phenomenon, and this is not part of it; mixing the two produces a term structure with a spurious kink at the front.

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The effect has a practical corollary that matters more than the misreading. A 0DTE option cannot hedge an overnight gap, because the gap happens after it has expired. An institution using daily options to hedge event risk must hold a contract expiring after the event, and for an announcement made after the close that means tomorrow's contract, not today's.

23.6 THE SMILE AT ZERO MATURITY

Section 21.12 said that this chapter's market lives entirely in the region where classical stochastic volatility cannot produce a skew, and Section 21.7 gave the empirical law: the at-the-money skew grows like $T^{-0.4}$ as maturity shortens, measured from about a month to a few years. Whether it continues below a month is disputed; rough volatility (Section 21.8) says it does. Extrapolated to this market, the law is dramatic.

Maturity	Skew, relative to 30 days	Volatility points per 1% of moneyness
30 days	1.00	2.5
7 days	1.79	4.5
1 day	3.90	9.7
1 hour	8.24	20.6

A one-day smile is roughly four times as steep as the 30-day smile, and an hour before expiration it is eight times as steep. The ratios are the law's; the 30-day level of 2.5 points is a stylized anchor, several times the one-month slope of Table 19.3's surface (about 0.4 points per 1%), and every slope below scales with it. The slope is negative, so it is the puts that carry it: a put one percent below spot can show an implied volatility twenty points above the at-the-money level. The same slope cannot be run the other way for a call at the same distance (twenty points below a 15% at-the-money volatility is not a volatility), so the call wing flattens and turns up instead, and the smile at this horizon is a steep put wing against a shallow, convex call wing. Figure 23.4 draws the four maturities on one axis: from a month to an hour the put wing goes from a gentle slope to a nearly vertical one, while the call wing moves by a few points.

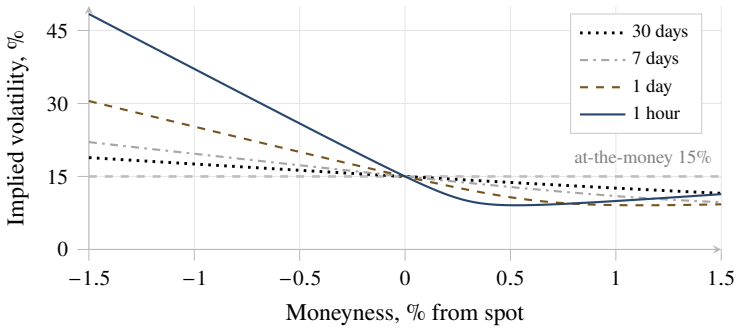


Figure 23.4: Implied volatility by moneyness at four maturities. Schematic: the at-the-money slope follows the $T^{-0.4}$ law of the table, extrapolated below one month, the at-the-money level is held at 15%, and one stylized wing shape is used throughout.

Three consequences follow, and the third is the one that matters for market structure.

The Black–Scholes implied volatility becomes a poor coordinate. Sections 15.7 and 19.12 described implied volatility as a language for quoting prices. At these maturities the language distorts: small differences in price map to enormous differences in implied volatility, and quoting in volatility terms conveys less information than quoting the price.

Only a model with jumps, roughness or very fast mean reversion can fit it. Example 21.3 showed jumps producing this shape (a smile whose range across strikes is 12.7 volatility

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points at one month and 1.5 at one year, almost all of it in the put wing), and Section 21.8 showed rough volatility producing the exponent directly. Whatever generates the 0DTE smile is not a classical diffusion with smoothly varying volatility, and this is the sharpest available evidence for that claim, because at these horizons such a path cannot travel far enough to give an out-of-the-money option any value; it takes jumps, very rough volatility, or a factor mean-reverting fast enough to act like either (Section 21.8).

And the steepness is where the retail flow is. The out-of-the-money puts that an intraday buyer of protection, or of a bet on a fall, naturally reaches for are precisely the strikes whose implied volatilities are most extreme. A buyer paying twenty points of skew is not making an obviously bad trade, since the steepness reflects a real probability of a downward jump, but they are buying the most expensive part of the surface, measured in the units the market itself uses. The call buyer at the same distance is on the cheap side of the same surface, paying less than the at-the-money volatility.

23.7 DEALER POSITIONING AND THE FEEDBACK LOOP

Section 18.12 established the mechanism, and the gamma term of equation (18.1) is its source: a move ΔS changes the dealer's delta by $\Gamma\,\Delta S$, and rehedging it trades with the move when dealers are *short* gamma, amplifying it, and against it when they are *long*, damping it. Applying that to 0DTE requires knowing which side dealers are on, and here the market's structure matters more than any model.

Retail buys options. The retail flow that Section 10.12 described is predominantly premium-paying: buying calls and puts outright. Against that flow dealers are short options and short gamma, and the feedback is amplifying.

But institutions sell them. A large and growing share of 0DTE volume is systematic premium selling—put spreads, call spreads, and iron condors (Section 12.3: a put spread below spot and a call spread above it, sold together) sold by funds harvesting the premium of Section 19.10 at the shortest available horizon. Against that flow dealers are long options and long gamma, and the feedback is dampening.

The two roughly offset, and the net is small and changes sign. This is the single most important empirical fact about the market and the one most often ignored. Studies of 0DTE order flow (Dim, Eraker, and Vilkov among them; see Further Reading) generally find it close to **two-sided**: customers buy options from dealers and sell options to dealers in roughly equal amounts, so dealers' net position is the difference of two large gross flows, and net dealer gamma from 0DTE contracts comes out a small fraction of that from longer-dated positions.

Example 23.3 How large the hedging flow is

Desks state gamma as the notional of the index that must be traded to rehedge a 1% move. A 1% move in the index changes the delta of one option by $0.01\,S\,\Gamma$ index units, so a position of n contracts with multiplier m needs

$$\text{hedge notional per } 1\% = 0.01\,n\,m\,\Gamma S^2$$

of index traded to restore the hedge. This is a different number from the gamma term of Table 23.1, the profit or loss on the same move, $\frac{1}{2}\Gamma S^2 \times 0.01^2$ per unit; the two describe one gamma, one as a trade and one as a P&L.

Suppose the dealer community is net short \$500 million of gamma per 1% from 0DTE positions. From Table 23.1 that is the gamma of about 930 at-the-money contracts an hour before expiration ($0.01 \times 100 \times 0.021528 \times 5000^2 = \$538{,}200$ each, the 100 being the multiplier m), or 13,000 at thirty days. Being short gamma, the dealers gain delta as the index falls and lose it as the index rises, so they must sell futures into a fall and buy into a rise; to first order, holding gamma fixed, the amount is \$500 million for each 1%:

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Index move	Index futures dealers must sell (fall) or buy (rise)
0.5%	$250m
1.0%	$500m
2.0%	$1,000m
3.0%	$1,500m

Each trade follows the move that caused it, which makes the flow amplifying. Gamma does not stay fixed at this horizon (Section 23.10): for the one-hour book the true figures are about $190m on a 0.5% move and stop near $230m, the whole of the book's delta, beyond 1%; for the thirty-day book the table is close.

Against S&P futures volume of several hundred billion dollars notional a day, these are small numbers in aggregate. A billion dollars of futures is minutes of normal volume.

They are not small at 3:55 p.m. New York time. The flow is concentrated in the final hour, by both charm and gamma, and within that window it can all be on one side. The close is the busiest part of the futures day, but a hedger who must sell now does not get to trade against the day's volume. They trade against what other participants are willing to buy at that moment at the best bid, which is a few hundred E-mini S&P 500 futures contracts (each $50 times the index, about $250,000 at 5000), or tens of millions to about a hundred million dollars. That is the right comparison, not daily volume, and against it a few hundred million dollars that all wants the same side is material.

23.8 JUMPS WITHIN THE DAY

Section 21.5 observed that jumps cluster across underlyings (the events that make one stock gap tend to make others gap) and left the intraday side to this chapter. For 0DTE the scheduled events behind that clustering are not a refinement—they are the dominant term.

Volatility within a session is not uniform. It concentrates at the open, at scheduled data releases, at policy announcements, and into the close. For a monthly option this averages away; for an option with six hours to live it does not. A 0DTE implied volatility is therefore not a number but a *schedule*, and a public one.

Example 23.4 What a two o'clock announcement does to a morning option

Take Section 23.5's session: 15% close-to-close volatility, 20% of variance overnight, so an ordinary session's standard deviation is 0.845% and a 0DTE option shows 13.4% implied volatility.

Assume ordinary variance is spread evenly through the session. Now add a scheduled announcement whose move has standard deviation s_e. Variances add, so the session's standard deviation becomes $\sqrt{0.845\%^2 + s_e^2}$; the last column is the implied volatility as a multiple of the 13.4% with no event:

Event move (s.d.)	Session s.d.	Implied volatility shown	Multiple
none	0.845%	13.4%	—
0.50%	0.982%	15.6%	1.16
0.75%	1.130%	17.9%	1.34
1.00%	1.309%	20.8%	1.55
1.50%	1.722%	27.3%	2.04

A single scheduled event with a one-percent standard deviation raises the morning's 0DTE implied volatility by more than half. Nothing about the market's ordinary volatility has changed; one known future minute has been added to the measurement window.

The sharper effect is what happens when the minute passes. Compare the option at 1:59 p.m. and at 2:01 p.m., with two hours left, the index unchanged, and the 1.00% event of the table:

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	Remaining s.d.	Annualized
Just before the announcement	1.104%	31.6%
Just after	0.469%	13.4%

The implied volatility falls by a factor of 2.36 in two minutes. The event variance has simply left the window. Three points follow.

This is not a crash in volatility; it is the calendar. A trader who buys the option at 1:59 and sells it at 2:01 without a view on the announcement has bought the event and sold the aftermath, which is a coherent trade. A trader who buys at 1:59 hoping to profit from a rise in volatility has misunderstood the instrument.

The calendar makes the constant-volatility assumption untenable, and not as a modeling nicety. Section 15.1 listed constant volatility among the Black–Scholes assumptions and Chapter 21 relaxed it. At this horizon the failure is not that volatility is stochastic—it is that volatility is *scheduled*, and everyone has the schedule. Desks price 0DTE against a calendar of event weights rather than a single number, which is a different kind of model from anything in Chapter 21; Section 23.9 describes it.

Finally, the clustering that Section 21.5 attributed to jumps is visible here directly. The events that make one index gap are the same events (data, policy, geopolitics) that make others gap, so intraday jump risk is correlated across underlyings exactly when it is largest. A book short out-of-the-money 0DTE options on several indices is not diversified; it is one position.

23.9 PRICING A 0DTE OPTION

Every number so far has come from the Black–Scholes–Merton formula with a 15% volatility and a clock in trading hours, and for the Greeks that was the right tool: the shape of gamma, charm and decay does not depend on the details. A reader who has followed the chapter this far will have two questions the formula does not answer. Which options are actually being traded? And if constant volatility is untenable, what does a desk use instead?

Which strikes trade. Mostly they are not the at-the-money ones. Cboe's own tabulation of 2023 volume and the academic samples named in the Further Reading agree on the shape. The typical contract is near the money but out of it (roughly a quarter to one percent away, a delta between about 0.10 and 0.35), and puts and calls trade in roughly equal numbers, the puts reaching further from spot than the calls do. Far out-of-the-money strikes, one to two percent away on a quiet day, account for many contracts and very little premium, because they cost a few dollars at most, and cents by the afternoon; counted by premium rather than by contract, the center of the market moves closer to the money. The distribution also moves through the day: in the morning the traded strikes span about a percent either side of spot, and by the last hour nearly everything is within about half a percent, the one-hour band of Figure 23.1, because that is the only place where any time value is left. In-the-money options trade little. By put–call parity, an in-the-money call is the out-of-the-money put at the same strike plus a long forward position: its time value is the put's, and the rest is index exposure bought at an option's wider bid–ask spread. That time value drains away through the day: at the open a call half a percent in the money carries about 15 points of it, an hour before the close about 4, and with fifteen minutes left about 1, at the skewed volatility of Section 23.6. A trader who wants the exposure buys the futures contract, and one who wants the option buys the out-of-the-money side. Retail flow is mostly single-leg, an outright call or put opened and often closed within the session, while institutional flow is mostly spreads: a put spread half to one percent below spot, a call spread the same distance above, or both together as an iron condor, held to expiration. Section 23.13 says who these people are; the point here is that the market's pricing problem is concentrated in a narrow band of near-the-money strikes and a steep wing of puts just below it.

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Why one volatility will not do. The Black–Scholes–Merton model makes three assumptions that are tolerable for a monthly option and fail together at this horizon. Volatility is constant, so variance accrues evenly with the clock; but Section 23.5 showed that a fifth of the day's variance falls in a window the contract does not cover, and Example 23.4 showed a scheduled minute carrying several times as much variance as the rest of the afternoon. The price path is continuous, so a strike several standard deviations away is nearly worthless; but at one hour a strike one percent below spot is 2.7 standard deviations away, and Section 23.6 showed the market pricing it as though the distance were far smaller, because a jump can cover it in an instant. And the hedge is continuous and free, so the price is the cost of replication; but Table 23.1 put gamma at fourteen times its monthly value and Example 23.1 put a dime of bid–ask at 2.7% of the premium, so the replication cost is neither small nor well defined. Example 23.5 shows how large the resulting error is.

Example 23.5 What a flat volatility gets wrong an hour before the close

Price a put below spot in the chapter's setting (index 5000, 4% rate, 1.5% dividend yield), first with a flat 15% volatility and then with the volatility given by the at-the-money slope of Section 23.6, extended linearly: 15% plus 2.5 points per percent of moneyness at thirty days, and plus 20.6 points per percent at one hour.

Time left	Strike	Flat 15%	With the skew	Ratio
30 days	0.5% below spot	83.99	92.45 (at 16.2%)	1.1
30 days	1.0% below spot	73.30	89.90 (at 17.5%)	1.2
1 hour	0.5% below spot	0.75	3.72 (at 25.3%)	5.0
1 hour	1.0% below spot	0.02	2.75 (at 35.6%)	148

At thirty days the smile is a correction of ten or twenty percent to a price the flat model gets roughly right. At one hour it is the price. The flat model values the put one percent below spot at two cents, because under a continuous path with a 0.37% hourly standard deviation the index almost never gets there; the market values it at about 2.75, over a hundred times as much, because a jump can. A desk that priced this put with the at-the-money volatility would offer it at the minimum tick and hedge it as if it were worthless.

What desks use instead. There is no one replacement; practice is closer to Chapter 19's than to Chapter 21's: the surface is the model, with the pieces assembled around it.

The first piece is the clock. Time to expiration is measured not in minutes but in expected variance remaining: an intraday profile heavy at the open and into the close with a thin middle, plus the weights of the day's scheduled events. This is **business time**, and the Black–Scholes–Merton formula is used with $\sigma^2 T$ replaced by that variance budget. It is why the option of Example 23.4 lost more than half its implied volatility in two minutes with nothing else changing: the budget was spent, not the volatility revised.

The second is the forward, which comes from the E-mini futures contract, the instrument the position will be hedged with, or from put–call parity within the chain as the VIX construction of Section 22.6 does. The smile is quoted against that forward, not against the cash index.

The third is the smile itself, fitted expiration by expiration in the standardized moneyness of Section 19.1,

$$\frac{\ln(K/F)}{\sigma\sqrt{T}},$$

the strike's distance from the forward in standard deviations, with σ the at-the-money implied volatility and T the business time remaining. The form is low-parameter and arbitrage-free by construction (the SVI parameterization of Section 19.5 is common), refitted through the day. In raw moneyness the smile steepens sharply as expiration approaches; in standardized

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coordinates most of that steepening disappears, and what is left, a slope proportional to $T^{0.1}$ (the $T^{-0.4}$ times the $\sqrt{T}$ the standardization divides by), is the residual the fit has to carry. The wings are where a view on jumps enters: a desk that expects an event to be larger than the calendar weight implies lifts the put wing directly, rather than changing a parameter in a dynamic model. The models of Chapter 21 are used off the desk, to check that the fitted surface is one some process could have produced. Jump models give a short-end skew that overshoots the market's, as Section 21.7 found, and rough volatility gives the $T^{-0.4}$ the market shows, but they do not set the quote.

The last piece is inventory. The spread and the skew adjustment around the fitted midpoint are set by the dealer's own position and by customers' orders: quotes are shaded against the book's gamma and charm strike by strike, widen into the close, and move with the flow. What the model supplies is the hedge ratios that make that possible, and for an actively traded 0DTE strike that is its whole job. The price is set by the bids and offers already in the market, not by anyone's formula.

23.10 PINNING, EVERY DAY

Section 10.9 introduced **pinning** alongside pin risk at the monthly expiration: the underlying is drawn toward a strike with large open interest, because dealers hedging a long gamma position sell into rallies and buy into declines around it. With a daily expiration, that mechanism now runs every session. Example 23.6 computes the size of the restoring force, because it is far larger than the monthly intuition suggests, and Figure 23.5 plots it.

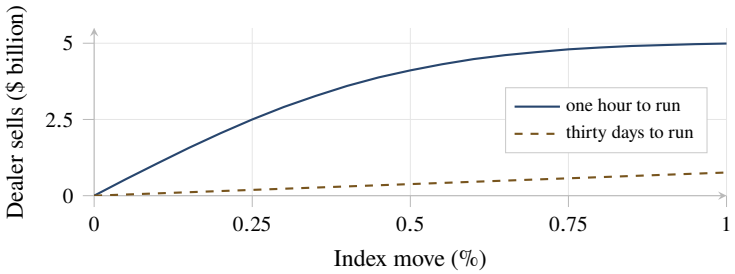


Figure 23.5: Dealer sale against index move for 20,000 calls held long at the money.

Example 23.6 The restoring force at one hour

Dealers are long 20,000 calls at one strike, the index sits at it, and one hour remains.

Index move	Delta goes from	Dealer sells
+0.10%	0.502 → 0.609	$1,063m
+0.25%	0.502 → 0.752	$2,499m
+0.50%	0.502 → 0.912	$4,114m
+1.00%	0.502 → 0.996	$4,990m

With thirty days to run the same book sells $382 million on a 0.5% move and $760 million on a 1% move, the straight line of Figure 23.5: doubling the move doubles the sale, which the one-hour curve plainly does not. Two features of the one-hour figures matter and they point in opposite directions.

Small moves meet an enormous restoring force. A tenth of a percent generates a billion dollars of stabilizing supply. This is genuine pinning, and it is why the index often trades in a narrow range near a strike with large open interest.

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But the force saturates. By a 1% move the delta has reached 0.996 and there is nothing left to sell: the last half-percent of the move generated only $876 million of additional hedging against $4.1 billion for the first half. The stabilizer is strongest for moves that did not need stabilizing and absent for the ones that did.

That asymmetry is the defensible form of the worry in Section 23.14. A market that suppresses small fluctuations and cannot suppress large ones does not merely fail to help in a crisis—it makes the crisis look more surprising, because the small moves that would ordinarily have signaled building stress were damped away.

23.11 MEASURING DEALER GAMMA, AND WHY THE NUMBER IS UNRELIABLE

Everything in Section 23.7 depends on the sign of net dealer gamma, and that quantity is not observable. The standard construction, which Section 18.12's case study introduced, takes the open interest OI_i of each listed contract i (a call or a put at some strike), computes its gamma Γ_i, and assigns the dealers a side, $s_i = +1$ (long) or -1 (short). The usual rule assumes them long all call open interest and short all put open interest. The published "gamma exposure," in the form SqueezeMetrics (2017) first set out, is then $\sum_i s_i \,\mathrm{OI}_i \, m \, \Gamma_i \, S^2 \times 0.01$, with m the contract multiplier (100 for SPX): the hedge notional per 1% of Example 23.3 (Section 18.7's gamma in dollars) with the signs assumed, and the "flip level" is the index level at which it changes sign.

The trouble is that the sign assumption is asserted, not measured. Commercial versions infer the side from trade data rather than assume it, which narrows the problem without removing it; Example 23.7 shows how much of the answer the assumption decides.

Example 23.7 The assumption is the answer

Take 100,000 calls and 100,000 puts outstanding at the same strike, with identical gamma per contract. Assume dealers are on one side of every contract. Let w_c be the fraction of the calls dealers are long (they are short the rest) and w_p the fraction of the puts they are short (they are long the rest), so that the standard rule is $w_c = w_p = 100\%$. Net dealer gamma is then 200,000 $(w_c - w_p)$ contracts' worth.

Assumed split (calls long / puts short)	Net dealer gamma (thousands of contracts)
100% / 0%	+200
80% / 20%	+120
60% / 40%	+40
50% / 50%	0
40% / 60%	−40
100% / 100%	0

The sign flips between the 60/40 and the 40/60 assumption, and nothing in public data determines which is right. The last row is the standard rule itself: applied to equal call and put open interest it returns exactly zero, so here the published number would be zero not because dealers are flat but because the assumption is symmetric. Open interest records that a contract exists; it does not record who is on which side.

A published gamma-exposure level is therefore what dealer positioning would be if the assumed split were true, a weaker statement than it usually receives. Three points follow:

The assumption is worst exactly where it matters. It is roughly defensible for a market dominated by call overwriting (investors selling calls against stock they hold) and institutional put buying. Section 23.7 argued that 0DTE flow is close to two-sided, which is the case in which the naive split is furthest from the truth.

The flip level inherits the error. A published "gamma flip" at some index level is a function of the assumed signs, so two analysts with different assumptions produce different flip levels from

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identical data.

None of this means the concept is useless. Dealer gamma is real, and Sections 18.12 and 23.10 depend on it; published estimates simply measure open interest precisely and positioning by assumption.

23.12 DOES 0DTE AMPLIFY OR DAMPEN?

This is the question the market cares about, and the answer is that the evidence is mixed and the sample is short.

The case that it amplifies. The mechanism of Section 23.7 is real. Gamma is enormous and concentrated in a narrow band, so hedging demand is a step function of the index level. Charm produces a large one-directional flow into the close. And the positions are concentrated and the hedging mechanical, which is the pattern that produced 1987 (Section 40.2), February 2018 (Section 22.8) and the snowball knock-ins of Section 35.7.

The case that it dampens. Short-dated option sellers, in aggregate, supply gamma to the market. A dealer long gamma sells into rallies and buys into declines, which is stabilizing. If systematic premium selling dominates, 0DTE makes the intraday market *less* volatile, not more.

What the data show. Realized intraday volatility on the S&P 500 has not risen in a way that tracks 0DTE growth (Dim, Eraker, and Vilkov 2024). There has been no episode in which 0DTE hedging has been convincingly identified as the cause of a large move, though two stress episodes have tested the market since 0DTE became its largest segment: August 5, 2024 and the April 2025 tariff sell-off, when VIX closed above 50. Accounts of both, the BIS's of August 5 among them, trace the moves to news and to leveraged positions being unwound, and on August 5 to wide pre-open option quotes that inflated the VIX itself, not to 0DTE hedging. And the aggregate net dealer gamma attributable to 0DTE appears small relative to the longer-dated book. Against this, Brogaard, Han, and Won find a rise in volatility and attribute it to speculative retail trading rather than to dealer hedging.

So the fair summary is: the mechanism exists and it has not yet produced an incident. Four reasons explain the absence. The flow is two-sided, the main reason and the most fragile. Expiration is a natural stop-loss, since tomorrow the position does not exist. Margin is collected daily and the positions are cleared: the 2018 episode ran through an exchange-traded note, XIV, whose prospectus allowed early termination (Section 22.8), while a listed option position is margined by the clearing house of Chapter 2 against a position that self-liquidates. And the underlying is the deepest equity market in the world. The last three are structural rather than behavioral, which, by the argument of Section 22.5's case study, is the kind that holds.

That summary deserves caution. *The stabilizing property is a property of the flow, not of the instrument.* Nothing guarantees that 0DTE order flow stays two-sided. It is two-sided because two groups happen to want opposite things at the same horizon, and either could stop.

Positions do not accumulate, which is genuinely reassuring. A 0DTE book is gone by the close. Compare Section 22.8's inverse products, whose short position was carried from day to day for years, or the snowball books of Chapter 35, whose knock-in exposure grew as the market fell over months. Whatever risk 0DTE creates is created and destroyed within one session, which puts a hard bound on how large it can get.

But the bound is on size, not on speed. A structure that resets daily cannot build a slow-motion catastrophe; it can still produce a violent hour. The events that would test this market (a large gap that arrives while dealers are short gamma near the close) are rare.

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Case Study 23.1 The same mechanism in a shallow market

The argument of Section 23.12 rests heavily on the depth of the S&P 500. What the identical mechanism does without that condition is documented in a 2021 SEC staff report.

In January 2021 a heavily shorted small-capitalization US stock attracted very large retail call buying, much of it in short-dated, far out-of-the-money strikes. The dealers who sold those calls were short gamma, and Section 23.7's arithmetic applied unchanged: as the stock rose, their delta grew, and they had to buy stock to stay hedged.

Everything is the same as the index case except one number. The S&P 500 absorbs a billion dollars of hedging in minutes. A stock with a small free float, a large fraction of which was already lent out to short sellers, could not absorb the equivalent at all. The hedging demand was large relative to the shares actually available, so it moved the price, which increased the hedging demand, which moved the price again. Three points need separating, since the episode is usually told as one about intent.

The dealers were not speculating. They were hedging a position they had sold, and a hedge that moves the thing it hedges is a structural feature of delta hedging in a market too small for it.

The short interest (shares sold short and not yet bought back) mattered as much as the options. Short sellers covering are buyers too, and the two flows reinforced one another. Attributing the whole move to option hedging overstates a mechanism that was real but not alone. That is the same caution Section 23.12 applies to 0DTE.

And the binding constraint was free float, not leverage. The options were not unusually leveraged; the underlying was unusually small, and that is the variable that decides whether the feedback of Section 23.7 is a curiosity or an event.

The lesson is therefore not that 0DTE is safe. The mechanism is settled, and the outcome depends on the ratio of hedging demand to available liquidity, which is small for the S&P 500 and was not for a single small-cap stock. To ask whether 0DTE is dangerous is to ask whether that ratio can change, and Section 23.14 lists the ways it could.

23.13 WHO TRADES 0DTE, AND WHY

Retail directional traders buy 0DTE options as a leveraged intraday bet. Example 23.1 gives the economics: a modest negative expected return, delivered as a rapid sequence of complete wins and complete losses.

Systematic premium sellers sell 0DTE spreads to harvest the volatility risk premium at the shortest horizon, where the annualized premium looks largest and, crucially, where a position cannot be carried into a multi-day drawdown. Selling a defined-risk spread that expires in hours is a different risk from selling a monthly straddle: the loss is bounded and the position cannot be held past its worst moment.

Institutional hedgers use 0DTE tactically around events within the session (a data release, a policy decision), and buy the next day's expiration for one announced after the close, where the risk is genuinely one-day and paying for a month of protection is waste. This is the least remarked and possibly the largest legitimate use, and it is a straightforward improvement in market completeness: a one-day risk can now be hedged with a one-day instrument.

Dealers and market makers intermediate, and their positioning is the transmission mechanism of Section 23.7.

The instrument itself is neutral: Section 12.8's replication argument, run across expiration dates instead of strikes, says a finer set of maturities strictly enlarges the space of achievable payoffs. What is new is the concentration of hedging it induces, which should be assessed

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separately.

23.14 WHAT WOULD HAVE TO GO WRONG

The useful form of the question is not "is 0DTE dangerous" but "what configuration would make it so," because that is checkable in advance.

One-sided flow. If the systematic sellers withdrew (after a bad quarter, or a capital charge, or a change in fund mandates), the retail buying would leave dealers uniformly short gamma.

A gap into concentrated gamma. The dangerous configuration is a large move arriving in the last hour, when the gamma band of Figure 23.1 is at its narrowest and the charm flow of Example 23.2 is at its fastest. A 1% move then takes the index outside the band, and the hedging that would have absorbed it runs out partway through, as Example 23.6 showed.

Liquidity that is not there. All of the numbers in Example 23.3 are small against average volume and large against the depth available in a fast market. The mechanism that matters is the one Section 2.9 identified in margin procyclicality: in a fast market the participants who would provide liquidity are the same ones who suddenly need it. And short-dated hedging is reaching shallower markets: since January 2026 stocks with a market value above $700 billion also carry Monday and Wednesday expirations (Section 10.2), which is the setting of Section 23.12's case study rather than the index's.

None of this is a prediction. It is a specification of what to watch, and the watchable quantities (the open-interest distribution near spot, the share of volume that is premium-selling, and estimates of dealer gamma by strike, read with Section 23.11's caution) are published or computable. The risk is not hidden, and unlike the episodes of Sections 22.8, 35.7 and 40.2, the profession is watching this one while it is small.

SUMMARY

Same-day options went from a single-digit share of SPX option volume to more than half in under a decade. Volume overstates the risk transferred; it does not overstate the hedging, which is compressed into hours.

Everything peculiar follows from the Greeks as $T \to 0$. Gamma an hour from expiration is many times gamma at thirty days, and the gamma band collapses to a narrow range around spot, so hedging demand becomes a step function of the index level. Charm becomes the dominant flow: a dealer short calls struck just above spot unwinds a large hedge in a day the index does not move, one way, predictably, fastest in the final hour. The buyer of an at-the-money call pays well under one standard deviation of the day's move and has lost most of the premium by the last hour if nothing happens.

Three things are routinely misread. A 0DTE option covers the session but is priced with a full day, so it shows less implied volatility than the daily figure and cannot hedge an overnight gap. Extrapolated below a month, where the evidence is disputed, the $T^{-0.4}$ law steepens the skew by nearly an order of magnitude from thirty days to the last hour before expiration. And volatility within the day is scheduled: an announcement lifts the morning's implied volatility, and the remaining option's falls sharply when it passes. One volatility therefore cannot price the market: a put just below spot an hour before the close is worth almost nothing at a flat volatility and many times as much with the skew—so desks price on a business-time clock against a smile fitted in standardized moneyness, and the model supplies hedge ratios, not the price.

Whether hedging amplifies or dampens depends on the sign of dealer gamma, which is not observable: retail buying and systematic selling roughly offset, and published estimates measure open interest precisely and positioning by assumption. Pinning happens daily and saturates: the hedging flow is heaviest on a small move and spent by a larger one. There has been no incident because the flow runs in both directions, positions expire before they

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accumulate, margin is daily, and the underlying is deep; the first of these is the one most likely to change, and the configuration to watch is flow running one way meeting a large move in the final hour.

FURTHER READING

This is the newest topic in the book and the literature is accordingly unsettled; a reader should expect the empirical picture to have moved. Cboe's own research notes on 0DTE volume composition (2023) are the source for what is actually traded and by whom, and they separate contract counts from risk transfer, which most commentary does not. On order flow and dealer positioning, Brogaard, Han, and Won (2023, revised 2026) find that 0DTE trading raises the volatility of the underlying and attribute the effect mainly to speculative retail trading rather than to dealer hedging, Dim, Eraker, and Vilkov (2024) find market makers' net 0DTE gamma small and, on average, long, with calmer rather than wilder trading after it, and Beckmeyer, Branger, and Gayda (2023) document the retail side (who buys, at what strikes, and with what returns); together they are the evidence Section 23.12 calls mixed, and they disagree less about dealer hedging than about whether 0DTE trading moves volatility at all. Bandi, Fusari, and Renò (2023) is the pricing paper, with time-varying skewness and kurtosis and jumps for the wings. On the feedback mechanism itself, the literature on option hedging and market impact predates 0DTE by decades and applies unchanged, and Grossman (1988) and the Brady Commission report are the place to start.

PRACTICE QUESTIONS

23.1 Explain why 0DTE contract volume overstates the amount of risk being transferred.

23.2 State how at-the-money gamma behaves as $T \to 0$, give the ratio between one-hour and thirty-day gamma from Table 23.1, and explain why the narrowing of the gamma band matters more than the growth in gamma itself.

23.3 An at-the-money 0DTE call costs 0.38% of spot. State the move required to break even at expiration, compare it with a one-standard-deviation daily move, and explain why the bid–ask spread is a much larger percentage cost late in the day than early.

23.4 Define charm, explain why it is negligible for a monthly option and dominant for a 0DTE option, and use Example 23.2 to state three properties that distinguish charm hedging from gamma hedging.

23.5 The overnight period carries 25% of daily variance. Compute the implied volatility a 0DTE option will show in a market whose close-to-close volatility is 16%, explain why this is not a term-structure effect, and state which contract would hedge an announcement made after the close.

23.6 Using the $T^{-0.4}$ law, compute how much steeper a four-hour smile is than a 30-day smile, and explain why quoting a 15-minute option in implied volatility terms conveys less information than quoting its price.

23.7 Explain why the pinning force of Example 23.6 saturates, and why a market that damps small moves but not large ones may be worse than one that damps neither.

23.8 A published gamma-exposure figure assumes dealers are long all calls and short all puts. State what is measured and what is assumed, and explain when the assumption is least defensible.

23.9 A 0DTE option shows 13.4% implied volatility with no event scheduled. A 2 p.m. announcement adds a move of 0.8% standard deviation. Compute the morning implied

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volatility and its value just after 2 p.m., with two hours left, and explain why the fall is not a fall in volatility.

23.10 Explain why retail flow makes dealers short gamma and systematic premium selling makes them long. A dealer book is net short \$800m of gamma per 1%. Compute the hedging trade a 1.5% move requires, state its direction, and explain why the same flow can be negligible against daily volume and material against the depth at 3:55 p.m.

23.11 State the four reasons Section 23.12 gives for the absence of an incident, identify which is behavioral rather than structural, and explain why daily expiration bounds the size of the risk but not its speed.

23.12 Explain what happens to dealer hedging demand when a move takes the index outside the gamma band, and why this is the dangerous configuration.

FURTHER QUESTIONS

23.13 Section 23.2 reports the gamma band's upper edge closing in from +6.76% to +0.44%. Derive the band's edges as a function of σ and T, show why the band is symmetric in log-strike about a point slightly above spot rather than about spot, verify the three pairs of figures, and determine how the band would change if implied volatility doubled. Discuss what this implies about whether 0DTE is more or less dangerous in a high-volatility regime.

23.14 Example 23.2 finds a \$1.07bn hedge unwind with the index unmoved. Derive the charm flow between the open and the last half hour for a book with a given open-interest distribution across strikes, explain why over that window the flow is largest for strikes roughly half a daily standard deviation from spot rather than at the money, say where the at-the-money strike's flow goes instead, and discuss what this implies about where in the strike distribution to look for the largest predictable flow.

23.15 Reproduce the one-hour rows of Example 23.5, then price the same two puts in Merton's jump-diffusion of Section 21.5 with the diffusion volatility set to 15%, and find the jump intensity and mean jump size that reproduce the skewed prices. Comment on how many parameter pairs do so, and on what that says about calibrating a dynamic model to a single expiration.

23.16 Section 23.12 argues the market is benign mainly because the flow is two-sided, and that this is the fragile reason. Construct a model of the two groups (retail buyers and systematic sellers) in which the equilibrium is stable, then identify the parameter changes that would break it, and state what observable series would give warning.

23.17 Section 23.14 claims that the hedging that would absorb a move "runs out partway through" once the move takes the index outside the gamma band. Formalize this: write dealer hedging demand as a function of the index level for a book concentrated at one strike, show where it is increasing and where it vanishes, and assess whether a mechanism that absorbs small moves and not large ones is worse than one that absorbs neither.

23.18 Section 23.13 argues that a daily expiration is a genuine improvement in market completeness. Make the argument precise using Section 12.8's replication result, then construct the counter-argument that finer maturities can reduce welfare by inducing hedging concentration, and state what would have to be measured to decide between them.

23.19 Compare 0DTE with portfolio insurance (Section 40.2), snowball autocallables (Section 35.7) and inverse volatility products (Section 22.8) along the dimensions of crowding, mechanical hedging, accumulation, and observability. Explain which features 0DTE

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shares and which it does not, and use the comparison to argue either that the profession's current level of concern is about right or that it is misdirected.

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